

RSC Class 2

More on Supply and Demand, and Elasticity

Meng Hsuan “Rex” Hsieh

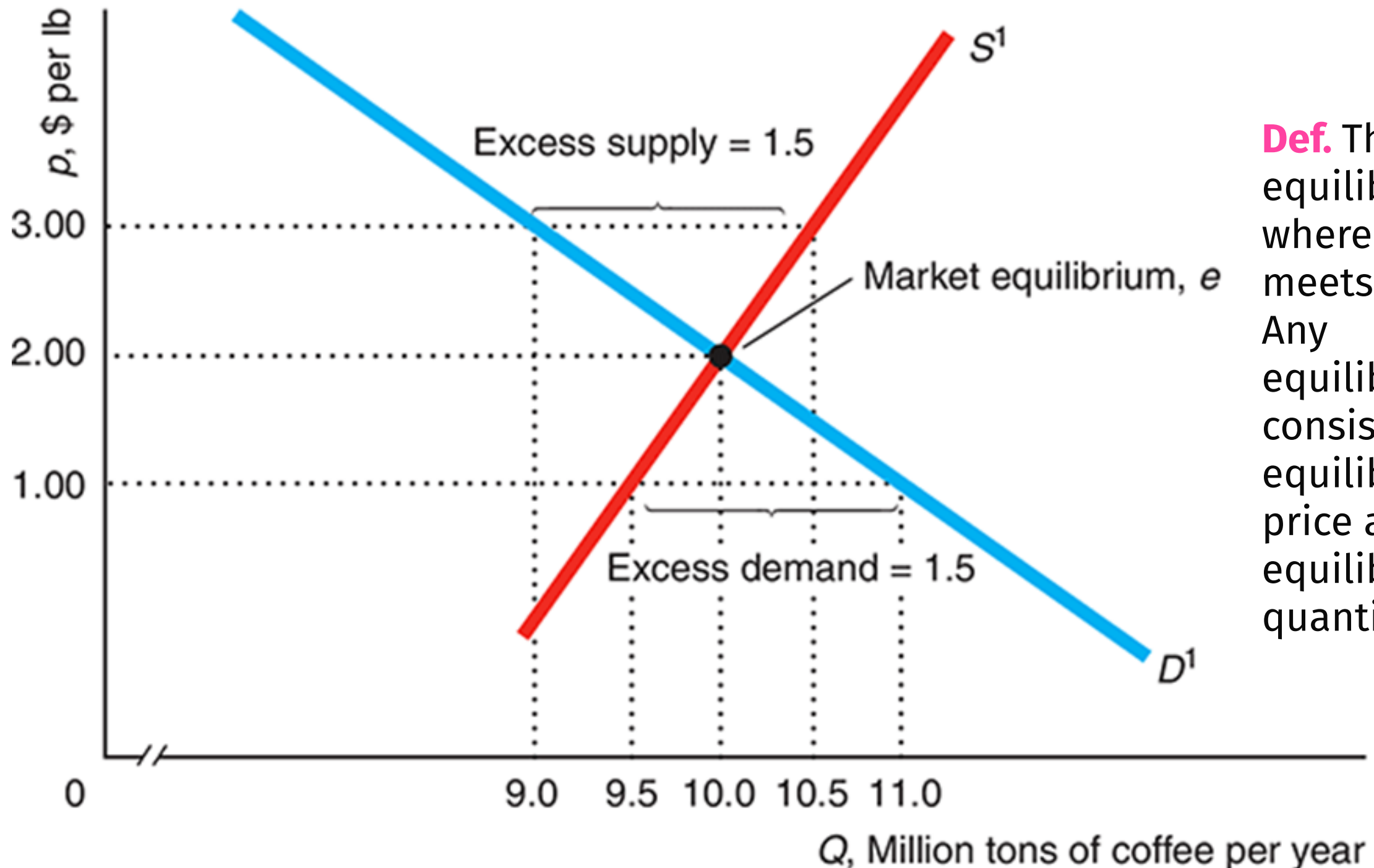
Agenda for Today

- More on supply/demand model
- Learning to do some calculations
- Elasticities and economic behavior
- ... Quiz Questions for Next Class!

More on Supply and Demand

Supply and Demand

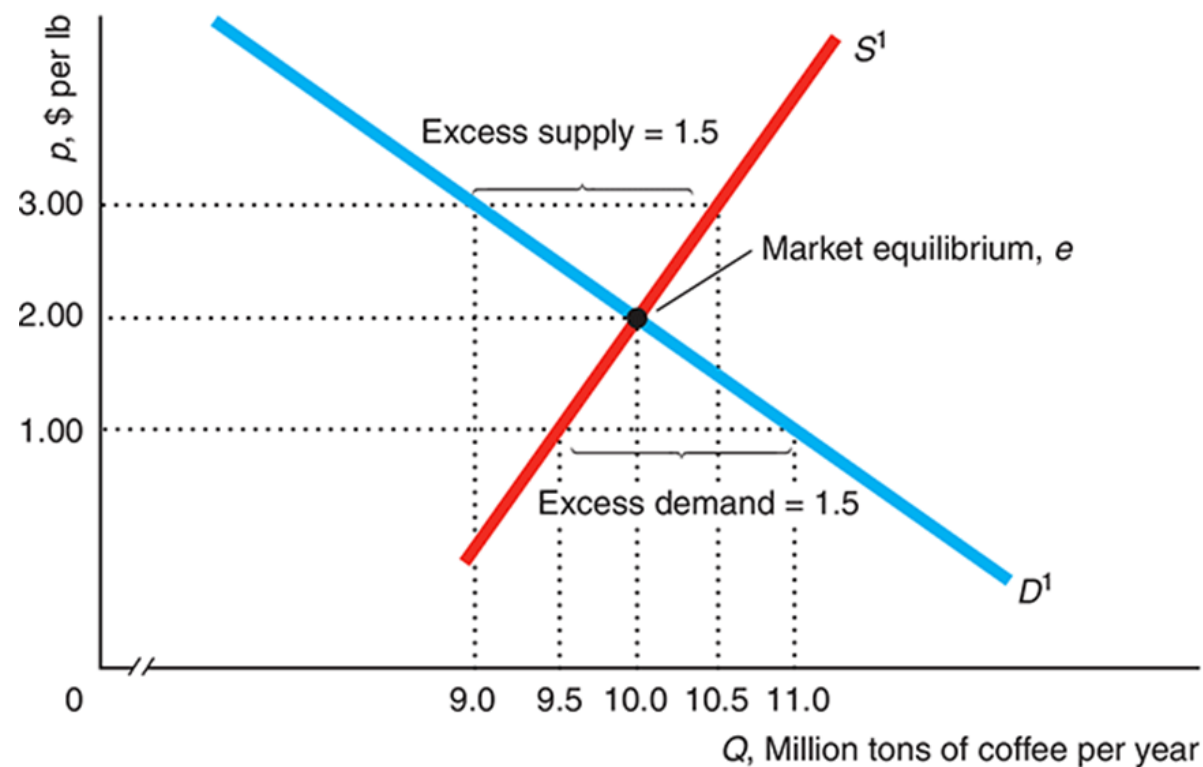
We left off here



Def. The market equilibrium is where supply meets demand. Any equilibrium consists of an equilibrium price and equilibrium quantity.

Supply and Demand

We left off here...



Remember markets are defined for a product, at a particular place, at a point in time

Therefore, any equilibrium describes a snapshot of individual consumers' and firms' behavior at a given time

What if there are fewer consumers?

What if all firms experience a cost increase due to tariffs?

Supply and Demand

We left off here... Movements along the supply/demand curves

- If only the price of a product goes up (everything else held constant), then a consumer would want to consume [more / less]
- Likewise, if only the price of a product a firm can sell at goes up (everything else held constant), then a producer is willing to produce [more / less]

These statements are equivalent to movement along the supply/demand curves

What if there are more than price changes?

Supply and Demand

We left off here... Shifts of the supply/demand curves

In life, there are many changes that are not related to price of the product changing

Example. Extreme climate events, limited supply of raw materials, shipping delays, shrinking customer base, ...

- If there are changes to any factors other than price that affect demand, then the entire demand curve shifts
- If there are changes to any factors other than price that affect supply, then the entire supply curve shifts

But be cautious about the market that we are analyzing

Supply and Demand

With algebra now

We can represent demand and supply using *demand functions* and *supply functions*, respectively

Example. $Q_D = 12 - P_D$ and $Q_S = 9 + 0.5P_S$ are (market) demand and (market) supply functions, respectively.

1. Find the market equilibrium quantity and price.

Supply and Demand

With algebra now

We can represent demand and supply using *demand functions* and *supply functions*, respectively

Example. $Q_D = 12 - P_D$ and $Q_S = 9 + 0.5P_S$ are (market) demand and (market) supply functions, respectively.

2. Suppose demand increased to $Q_D = 16 - P_D$, but supply remained the same. Without doing any calculations, the new market equilibrium quantity [increased / decreased] and the new market equilibrium price [increased / decreased] compared to those in question 1.

Supply and Demand

With algebra now

Any change to market condition is called a **shock**

Market equilibrium changes only if a shock occurs that shifts demand or supply curve

Demand or supply curve shifts occur if variables we were holding constant (tastes, income, government policies, costs of production change) change

Movement along the demand or supply curve are a result of only price changes

Supply and Demand

Let's return to excess supply and excess demand

Example (Continued). $Q_D = 12 - P_D$
and $Q_S = 9 + 0.5P_S$ are (market)
demand and (market) supply functions,
respectively.

3. Suppose the current market price is at $\bar{P} = 0.79$ (\$/pound). Is the market in equilibrium? If not, is there excess supply or demand?
4. Same as question 1: but what if the current market price is at $\bar{P} = 2.19$ (\$/pound) instead?
5. Explain why markets move toward equilibrium.

Supply and Demand

Let's return to excess supply and excess demand

Equilibrium captures a revision process in markets

We will assume markets are in equilibria unless otherwise specified

Supply and Demand

Simultaneous shifts in supply and demand

It is also completely reasonable to be in situations where both supply and demand shift, potentially in different directions

These have implications for market equilibrium quantity and price

Supply and Demand

Simultaneous shifts in supply and demand

In summary:

- S shifts out & D shifts out, $Q \uparrow$ and $P \uparrow \downarrow =$
- S shifts out & D shifts in, $Q \uparrow \downarrow =$ and $P \downarrow$
- S shifts in & D shifts out, $Q \uparrow \downarrow =$ and $P \uparrow$
- S shifts in & D contract, $Q \downarrow$ and $P \uparrow \downarrow =$

Exercise: draw these out!

Supply and Demand

With our understanding of supply and demand: we now know that each point on supply and demand curve describes a particular willingness-to-produce and willingness-to-pay

We can ask how we can quantify behavior at various points along the supply and demand

The right concept to use is called **elasticity**

Elasticity

Elasticity

... Is a statement about changes to behavior when price changes

BUSINESS • EARNINGS

Hasbro Says Higher Toy Prices Didn't Dent Holiday Demand

The maker of Play-Doh and Nerf said revenue in the fourth-quarter jumped to \$1.45 billion

By Connor Hart

Updated Feb. 10, 2026 12:17 pm ET

Hasbro HAS 0.16% ▲ said shoppers were willing to pay higher prices for toys during the holiday season, allowing the company to pass along tariff costs without significantly hurting demand.

Demand held up better than expected despite the higher prices, helping Hasbro swing to a fourth-quarter profit and post a 31% jump in revenue. Shares climbed 7.4% to \$103.95 after the company projected another gain in sales for the coming year.

Elasticity

Def. (Own-price) Elasticity of demand is the ratio of percentage change in quantity to the percentage change in price. This is also called point elasticity of demand

We define the **(own price) elasticity of demand** ε as

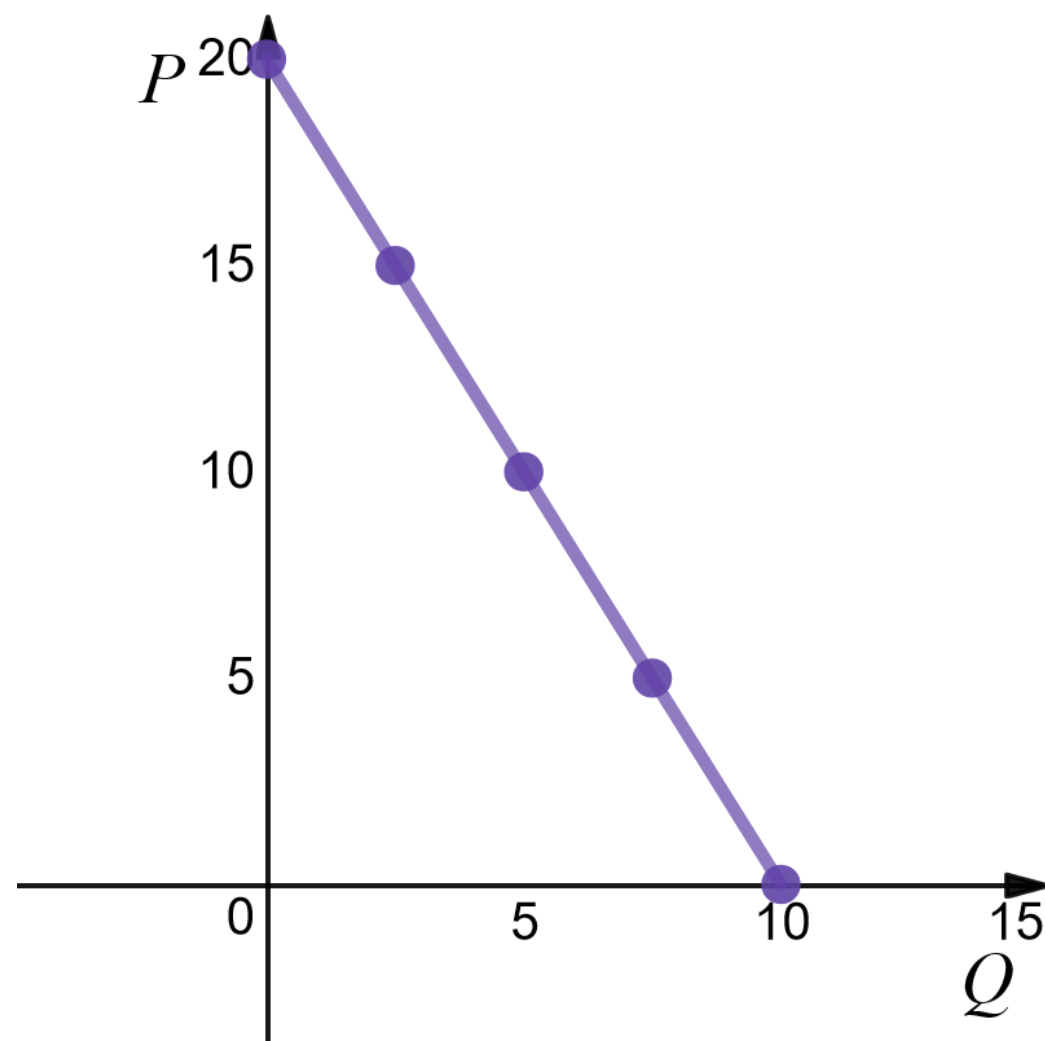
$$\varepsilon \stackrel{(1)}{=} \frac{\% \Delta Q}{\% \Delta P} \stackrel{(2)}{=} \frac{\Delta Q / Q}{\Delta P / P} = \frac{\Delta Q}{\Delta P} \frac{P}{Q}$$

$\Delta Q / \Delta P$ is slope of demand function

By definition, ε is negative, but we follow tradition and work with $|\varepsilon|$ as the elasticity of demand. **Let's unpack this definition**

Elasticity

In words, elasticities tell us how individuals' quantity demanded (or supplied) changes when price changes

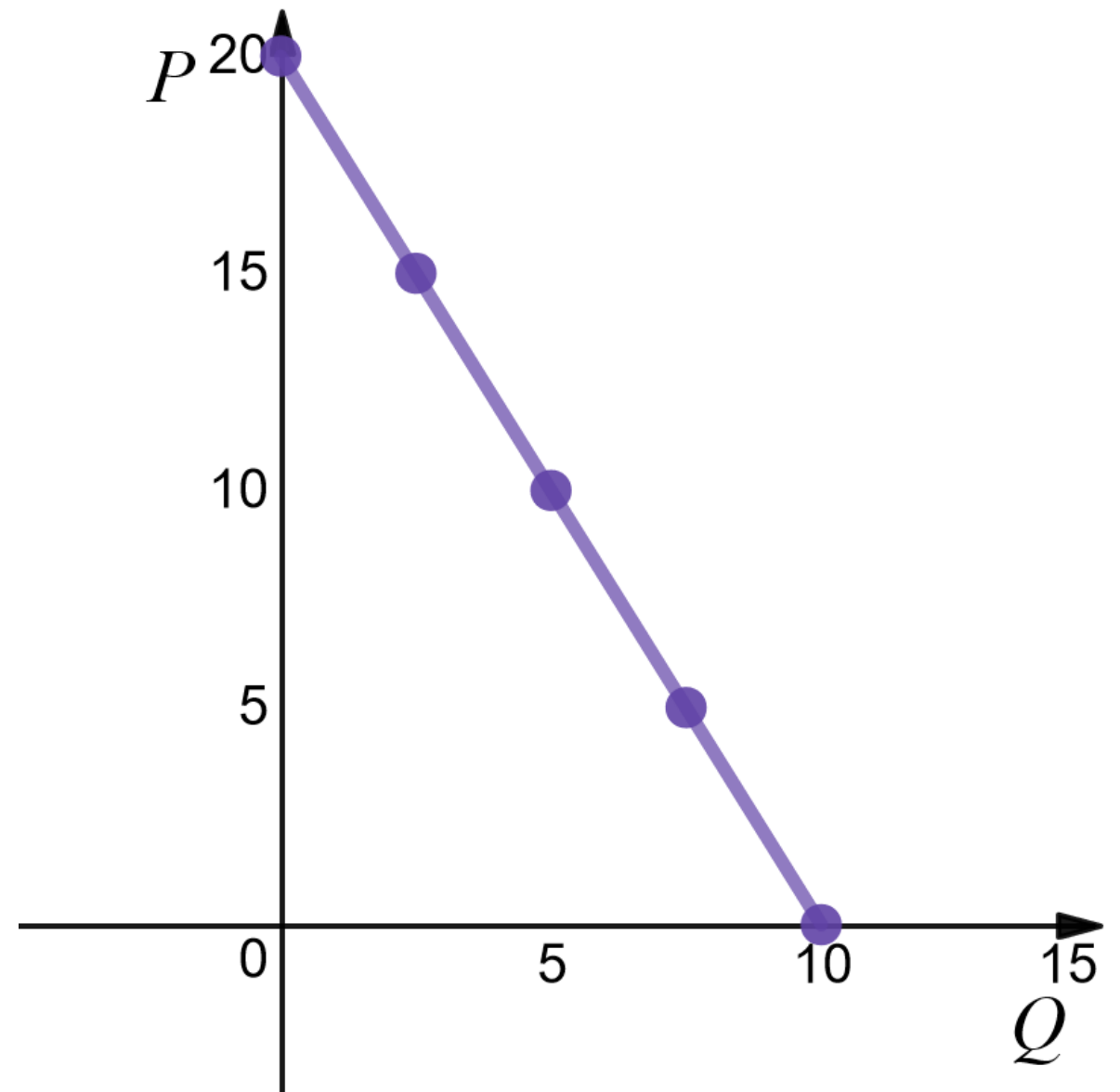


By definition, the point elasticity of demand changes along the demand (or supply) curve!

Example. Given the demand function $Q = 10 - 0.5P$ (illustrated on the left), calculate the elasticity of demand at $(5, 10)$, $(0, 20)$ and $(10, 0)$.

Elasticities with linear demand

- Elastic above midpoint
- Unit-elastic at midpoint
- Inelastic below midpoint



Elasticity

We can also say the demand (or supply) curve is elastic or inelastic

Def. Demand is (relatively) elastic if, for a given percentage change in price, the resulting percentage change in quantity demanded is large

Def. Demand is (relatively) inelastic if, for a given percentage change in price, the resulting percentage change in quantity demanded is small

Let's illustrate (relatively) elastic and inelastic demand on a graph

Elasticity

Elastic demand versus inelastic demand

Elasticity

Applying our understanding... to pre-class work!

WSJ ran the following article

BUSINESS | RETAIL | HEARD ON THE STREET [Follow](#)

Soaring Costs Expose a Trans-Atlantic Chocolate Divide

As cocoa prices surge, Europe's chocolate habit is proving more resilient than America's

By [David Wainer](#) [Follow](#)

May 30, 2025 at 5:30 am ET

[Share](#) [AA](#) [Resize](#)

[Listen](#) (2 min) [⋮](#)

Since 2021, chocolate prices have risen more than 30% in both the U.S. and Europe, according to TD Cowen estimates based on Nielsen data. But the impact on demand has been more pronounced in the U.S., due to key differences in price elasticity between the two markets. While companies with a strong European presence like Mondelez have been able to pass on price increases there without sacrificing volume too much, U.S.-centric manufacturers like Hershey have found North American consumers to be more fickle.

Almost everyone loves chocolate, whether it's a Hershey's Kiss, a Toblerone or a dark Lindt bar.

But a historic surge in cocoa prices is testing just how deep that love really runs. And it turns out Europeans may be a bit more committed to chocolate than their American counterparts. That gap in loyalty has real implications for global confectionary companies.

Yet Americans and Europeans tend to satisfy their sugar cravings in different ways, and Wall Street is paying close attention. American consumers are often more impulsive, picking up a bar as a last-minute decision at the checkout line or while filling up at a gas station, notes Alexia Howard, an analyst at Bernstein. In contrast, Europeans treat chocolate more like a daily staple. Bars with higher cocoa content are often a regular item on the family grocery list in Europe, which accounts for roughly 50% of global sales.

Hershey gets about 87% of its overall net sales from the U.S. Mondelez, meanwhile, gets 60% of its chocolate sales from Europe and another 35% from developing markets. Less than 5% of its chocolate business is in North America, though it does have a wider snacking business there.

For this year, Mondelez [is forecasting a 10% drop](#) in adjusted earnings per share, which is relatively mild when compared to a mid-30% estimated decline at Hershey. Over the past two years, Hershey's stock is down about 35%, while Mondelez has fallen roughly 5%.

Elasticity

Applying our understanding... to pre-class work!

WSJ ran the following article

BUSINESS | RETAIL | HEARD ON THE STREET [Follow](#)

Soaring Costs Expose a Trans-Atlantic Chocolate Divide

As cocoa prices surge, Europe's chocolate habit is proving more resilient than America's

By [David Wainer](#) [Follow](#)

May 30, 2025 at 5:30 am ET

[Share](#) [AA](#) [Resize](#)

[Listen](#) (2 min)

Since 2021, chocolate prices have risen more than 30% in both the U.S. and Europe, according to TD Cowen estimates based on Nielsen data. But the impact on demand has been more pronounced in the U.S., due to key differences in price elasticity between the two markets. While companies with a strong European presence like Mondelez have been able to pass on price increases there without sacrificing volume too much, U.S.-centric manufacturers like Hershey have found North American consumers to be more fickle.

Almost everyone loves chocolate, whether it's a Hershey's Kiss, a Toblerone or a dark Lindt bar.

But a historic surge in cocoa prices is testing just how deep that love really runs. And it turns out Europeans may be a bit more committed to chocolate than their American counterparts. That gap in loyalty has real implications for global confectionary companies.

Yet Americans and Europeans tend to satisfy their sugar cravings in different ways, and Wall Street is paying close attention. American consumers are often more impulsive, picking up a bar as a last-minute decision at the checkout line or while filling up at a gas station, notes Alexia Howard, an analyst at Bernstein. In contrast, Europeans treat chocolate more like a daily staple. Bars with higher cocoa content are often a regular item on the family grocery list in Europe, which accounts for roughly 50% of global sales.

What does this imply about the (relative) elasticities of demand for chocolate between Americans and Europeans?

Elasticity

Other elasticity measures

A few more pieces of terminology

Def. For any good X , if the income elasticity of demand for X is *negative* (*positive*), then we say X is *inferior* (*normal*) good.

Def. For any two goods X and Y , if the cross-price elasticity of demand for X with respect to Y is *negative* (*positive*), then we say X and Y are *complements* (*substitutes*).

Elasticity

Other elasticity measures

Examples.

1. Rex's net income increased by 21% this year compared to last; his purchases of sparkling water rose by 17%. What is his income elasticity of demand for sparkling water?

Elasticity

Other elasticity measures

Examples (continued).

2. Wicked Art Fair (WAF) and Artistic Odyssey (AO) hold fairs at the same time every year. But, this year, WAF raised their price by 20%, and AO sees their traffic increased by 11%. What is the cross-price elasticity of demand of (tickets for) AO?

Elasticity

With marginal revenue

We're now going to link elasticity to marginal revenue

The intellectual exercise is:

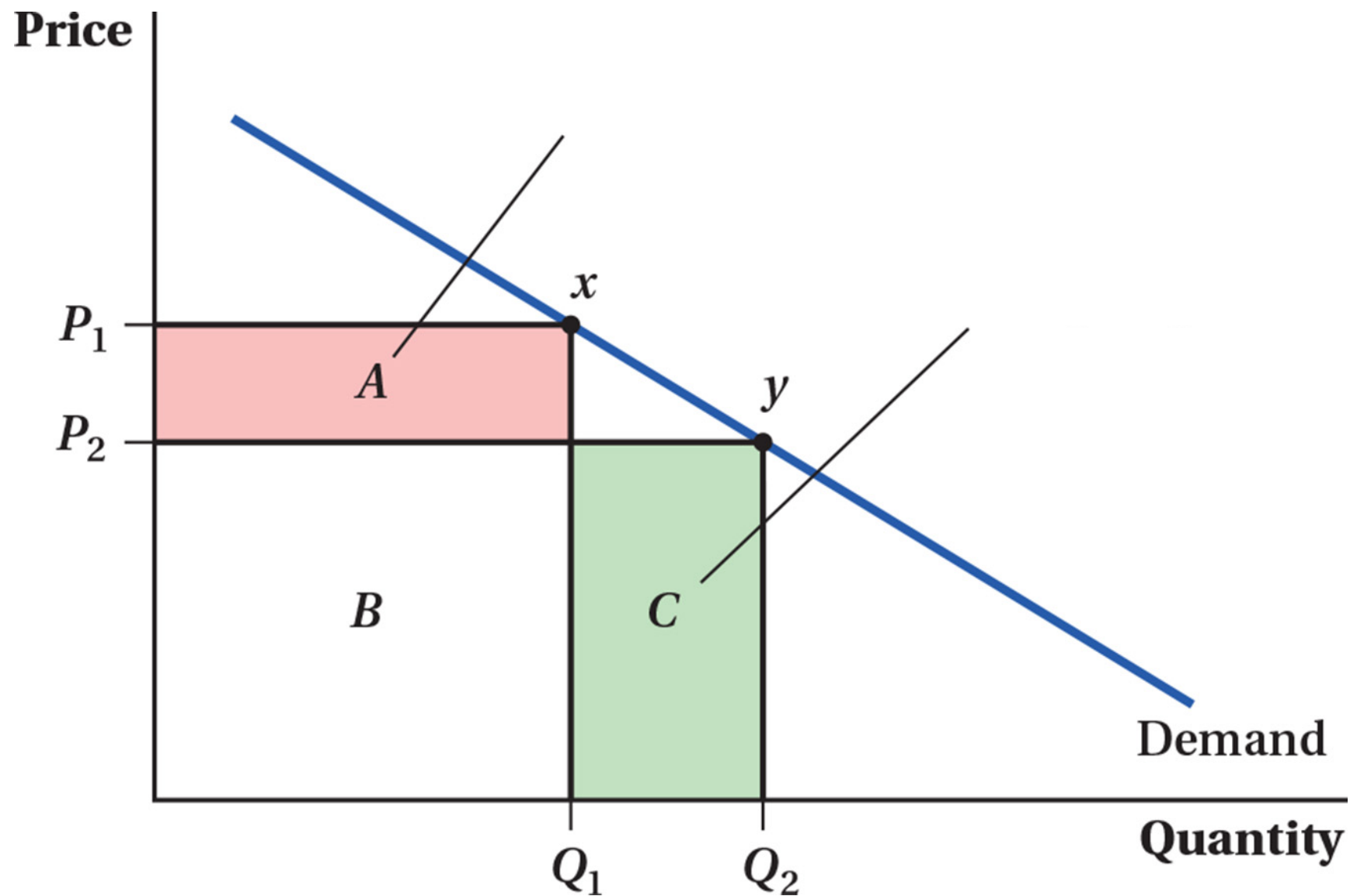
Imagine we own a firm, and we face a group of potential customers — each with the same demand for our product

How can I relate their behavior to incremental revenue we will as a firm?

You might ask: why should we, as a firm, care about this? And how would we even know the answer to this question?

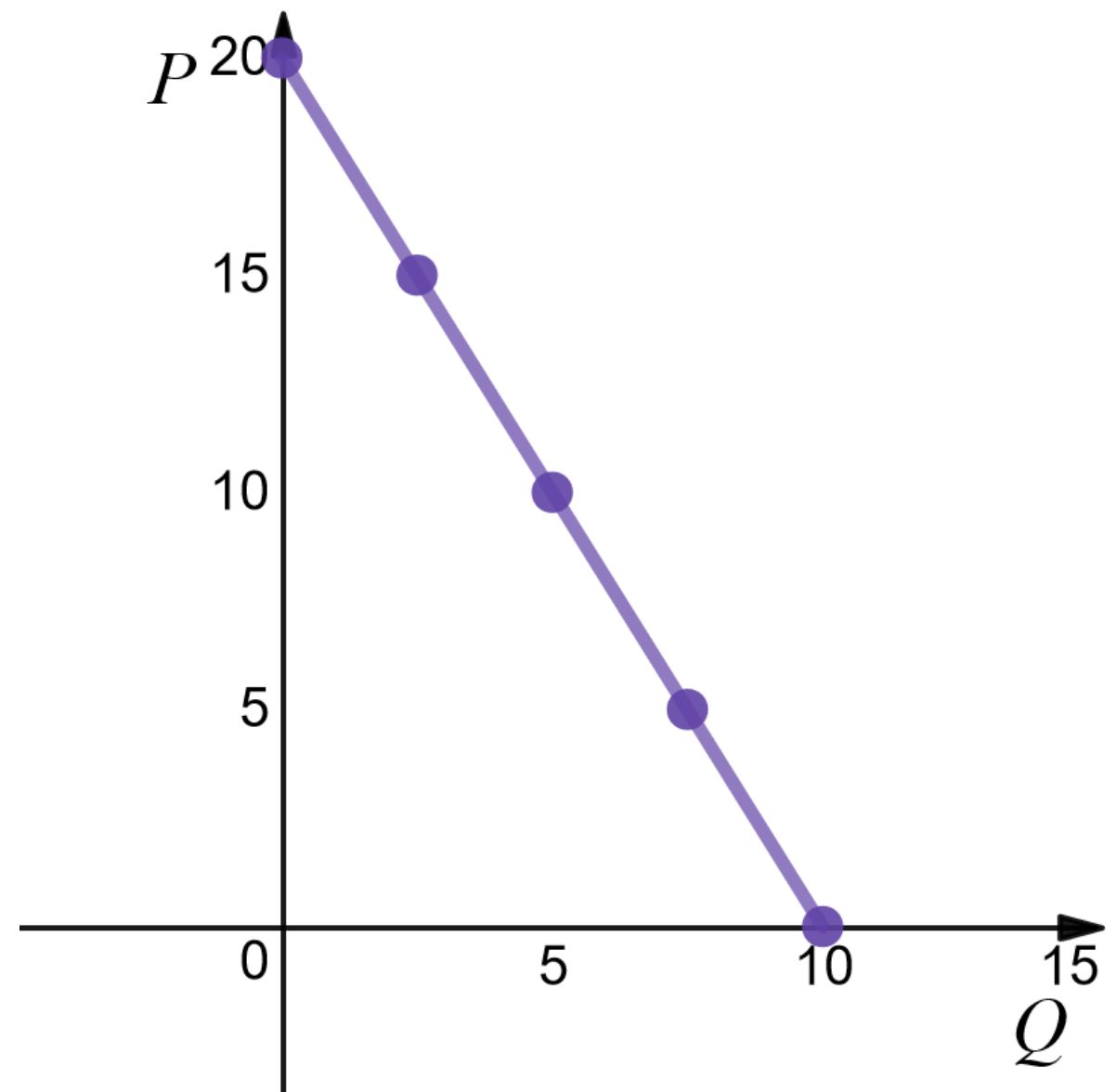
Elasticity

With marginal revenue



Elasticities with linear demand

- Elastic above midpoint
- Unit-elastic at midpoint
- Inelastic below midpoint



Fact: for a firm facing a given demand function, the firm's marginal revenue is zero when they set a price where the customer's demand is unit-elastic (at midpoint if we're working with linear demand)

This will our first foray into analyzing market behavior!

Elasticity

With marginal revenue; a preview of next class...

The fact from previous slide is a bit mysterious, let's unpack that

We will be using **marginal revenue** a lot going forward

This is the concept of marginal benefits for firms

Def. Marginal revenue is the incremental revenue made by firms when they produce one additional unit of product

Mathematically, marginal revenue is the *derivative* of the revenue function

For next time

We're gonna start moving quickly — come to office hours later this week w/ any questions/concerns

See you tomorrow (@ 2.45pm sharp)!